

KESHAV BHANDARI

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SUMMARY

I am a 2nd year PhD student in the Department of Computer Science at Queen Mary University of London. My research revolves around neuro-symbolic approaches for generating music with a focus on controllability and musical structure.

EDUCATION

Queen Mary University of London, School of Electronic Engineering and Computer Science PhD in Computer Science – Neuro Symbolic Automated Music Composition	Sep 2023 – Present London, UK
Northwestern University, McCormick School of Engineering Master of Science in Artificial Intelligence	Sep 2021 – Dec 2022 Evanston, IL, USA
Purdue University, Krannert School of Management Bachelor of Science in Business Management & Marketing (Dean's List, 2013)	Aug 2012 – May 2016 West Lafayette, IN

PUBLICATIONS

- Bhandari, K.**, Chang, S., Lu, T., Enus, F. R., Herremans, D., & Colton, S. (2024). ImprovNet: Generating Controllable Musical Improvisations with Iterative Corruption Refinement. Manuscript submitted for publication, IJCNN.
- Bhandari, K.**, Roy, A., Wang, K., Puri, G., Colton, S., & Herremans, D. (2024). Text2midi: Generating Symbolic Music from Captions. Proceedings of the AAAI Conference on Artificial Intelligence (AAAI-2025). (In press).
- Bhandari, K.**, & Colton, S. (2025 Submitted). Yin-Yang: Developing Motifs With Long-Term Structure And Controllability. In International Conference on Artificial Intelligence in Music, Sound, Art and Design (EvoMUSART 2025).
- Colton, S., Bradshaw, L., Banar, B., & **Bhandari, K.** (2024). Automatic Generation of Expressive Piano Miniatures. International Conference on Computational Creativity (ICCC).
- Bhandari, K.**, & Colton, S. (2024). Motifs, Phrases, and Beyond: The Modelling of Structure in Symbolic Music Generation. In International Conference on Artificial Intelligence in Music, Sound, Art and Design (EvoMUSART 2024).
- O'Reilly, P., Bugler, A., **Bhandari, K.**, Morrison, M., & Pardo, B. (2022). VoiceBlock: Privacy through Real-Time Adversarial Attacks with Audio-to-Audio Models. Advances in Neural Information Processing Systems, 35, 30058-30070.

CONFERENCE PRESENTATIONS

- “Towards Melodic Development with Discrete Diffusion Models for Symbolic Music”, DMRN+18: Digital Music Research Network One-Day Workshop 2023, Queen Mary University of London, UK, Dec’ 23.
- “Augmentations to improve rare bird call classification for a highly imbalanced multi-label soundscape environment”, 11th International Conference on Ecological Informatics (ICEI 2020+1), Digital University Kerala, India, Nov’ 21

GRANTS AWARDED

- AI for Earth grant, Azure, Microsoft, Oct’ 21, “Using deep learning to build a robust automated recognition system for rare and endangered birds in India”.

TEACHING

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| Demonstrator for Computational Creativity | Jan 2024 - Present |
| Senior Demonstrator for Data Science and AI master's cohort | Sep 2024 - Present |
| Demonstrator for ECS7024P - Statistics for Artificial Intelligence and Data Science | Sep 2024 – Dec 2024 |
| Teaching Fellow – Supervising 6 UG and 1 PG students on their projects | Oct 2024 - Present |
| Google DeepMind Research Ready Programme - PhD Mentor for two students on a research project. | Jun 2024 – Jul 2024 |

PROFESSIONAL EXPERIENCE

Boston Consulting Group – Data Scientist (Los Angeles, CA, USA)	Jan 2023 – Sep 2023
Vodafone Italy: Developed AI-based tools using Llama and Stable Diffusion to enhance digital marketing efficiency. Staples: Designed a pattern matching algorithm to improve control group selection for precise reporting and strategic insights. Walgreens: Built model drift scripts to rectify deviations in price optimization algorithm in a MLOps framework.	
Epsilon – Senior Data Scientist (Bangalore, Karnataka, India)	Sep 2017 – Feb 2021
- Built deep learning based personalized recommendation engines using Transformers, boosting client hit rates by up to 5%. - Deployed end-to-end ML & DL solutions as a product on cloud platforms, achieving lift rates between 3X to 9X. - Developed Gredel, an interactive R-Shiny package for automating predictive modeling and improving ML interpretability.	

ACCOMPLISHMENTS

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| 2 nd place in Epsilon India Recommendation Engine Hackathon (26 Teams) | 2020 |
| Top 6% Kaggle Categorical Feature Encoding Challenge (1,342 Teams) | 2019 |
| Champion Innovator Spot Award, Epsilon | 2019 |

- Top 3% Kaggle TalkingData Ad-Tracking Fraud Detection Challenge (3,946 Teams) 2018

LEADERSHIP & COMMUNITY SERVICE

Data Kind, Bangalore Chapter	<i>Aug 2018 – Oct 2019</i>
President, Purdue Marketing Association, Purdue University	<i>Jan 2015 - Dec 2015</i>